

Module 10: Error Detection: Parity, Checksum & Cyclic Redundancy Check (CRC)

Unit 2: Data Link Layer & MAC Sublayer

Error Control & Polynomial Mathematics

Mathematical foundations of transmission error detection: 2D parity, 1's complement Internet checksum, Modulo-2 polynomial division (CRC), and error catching proofs.

Error Detection: Cyclic Redundancy Check (CRC) Polynomial Division & Checksum

1. CRC Modulo-2 Binary Division (XOR Arithmetic)

Data: 100100 (6 bits) | Divisor G(x): 1101 (degree r = 3 zeros appended)

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Divisor: 1101 ) 100100 000 ( Appended r=3 zeros )
               1101
               ---
               01000 (XOR: 1^1=0, 0^1=1, 0^0=0, 1^1=0)
                1101
                ---
                01010
                 1101
                 ---
                 01110
                  1101
                  ---
                  0011 → Remainder CRC = 001
```

Transmitted Codeword = Data + CRC = 100100 001

2. Parity & Internet Checksum Methods

Two-Dimensional (2D) Parity:

Data organized into a matrix of rows and columns.

Adds row parity bits AND a column parity byte.

Detects all 1, 2, 3-bit errors! Corrects any single-bit error!

Internet Checksum (16-bit 1's Complement):

1. Split message into 16-bit words.

2. Add all words using 1's complement (wrap-around carry!).

3. Checksum = 1's complement of sum. Receiver sum must be all 1s.

Receiver Verification in CRC:

Receiver divides incoming codeword (100100001) by G(x) (1101).

If Remainder == 0 → ACCEPT! If Remainder != 0 → ERROR DETECTED!

Standard CRC Polynomials & Burst Error Detection Capabilities

CRC-16: $G(x) = x^{16} + x^{15} + x^2 + 1$

Used in HDLC and USB. Appends 16 check bits.

CRC-32: $G(x) = x^{32} + x^{26} + x^{23} + \dots + 1$

Standard in IEEE 802.3 Ethernet FCS! Detects 100% burst errors ≤ 32 bits!

Burst Error Definition:

Length of burst is measured from first corrupted bit to last corrupted bit (even if intermediate bits are intact).

CRC Detection Proof:

If polynomial has at least two terms: detects all single bits. If $(x+1)$ is a factor: detects all odd numbers of bit errors.

1. CRC Modulo-2 Arithmetic

Cyclic Redundancy Check relies on polynomial algebra over Galois Field GF(2). Bit addition and subtraction are performed via carry-less XOR operations. Appending r zeros followed by division yields the r -bit Frame Check Sequence (FCS).

CRC Operational Formula

Appended Message: $M(x) \cdot x^r$

$M(x) \cdot x^r = Q(x) \cdot G(x) + R(x)$

Transmitted Codeword $T(x) = M(x) \cdot x^r + R(x)$

Receiver Check: $T(x) \bmod G(x) == 0$ (Valid)